

FINAL PROJECT REPORT – SQA (DEMOQA)

1. Project Details

Project Title: Bug Tracking and Test Automation for DemoQA Web Application

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Course: Software Quality Assurance (SQA)

Duration: 3 Weeks

2. Project Overview

This project focuses on manual testing, bug reporting, test planning, and automation testing of the DemoQA web application. The objective was to understand real-world QA processes including SDLC, STLC, functional testing, regression testing, cross-browser testing, and automation using Selenium and Postman.

3. Testing Methodology

The following testing approaches were used throughout the project:

- Manual Testing (Week 1)
- Test Case Design and Bug Reporting
- Test Planning and Execution (Week 2)
- Black Box Testing (Equivalence Partitioning & Boundary Value Analysis)
- Cross-Browser Compatibility Testing
- Automation Testing using Selenium WebDriver (Week 3)
- API Testing using Postman (GET, POST, PUT, DELETE)

4. Modules Tested

- Forms (Practice Form)
- Web Tables
- Book Store Application
- Alerts, Frames & Windows (Automation)
- API Endpoints (Postman Testing)

5. Test Case Summary

- Total Test Cases Designed: 11 (Manual)

- Automation Coverage: 3 Test Cases automated using Selenium WebDriver (from existing manual test cases)
- Passed Test Cases: 7
- Failed Test Cases: 4

Key areas tested:

- Form validation
- Table operations (Add/Edit/Delete)
- Navigation and UI behavior
- Search functionality
- Cross-browser compatibility

6. Bug Report Summary

Major issues identified:

- Advertisement overlapping UI elements
- Form validation inconsistency
- Popup close button not responsive
- Web Tables performance issues
- UI alignment issues across browsers

All bugs were documented and retested in Week 2.

7. Automation Testing (Selenium)

Automation scripts were developed using Selenium WebDriver to validate key functionalities:

✓ Automated Test Cases:

- Form submission validation
- Web Tables Add/Edit functionality
- Navigation to modules
- Alerts handling (basic automation)

✓ Evidence:

Screenshots of:

- Script execution
- Browser automation results
- Terminal output

8. API Testing (Postman)

API testing was performed using Postman to validate backend responses.

✓ Methods Used:

- GET → Retrieve data
- POST → Create new record
- PUT → Update record
- DELETE → Remove record

✓ Observations:

- All endpoints returned expected status codes
- JSON responses were validated successfully

(Screenshots included in report)

9. Key Learnings

- Understanding of SDLC and STLC lifecycle
- Manual vs Automation testing difference
- Real-world bug tracking process
- Cross-browser testing challenges
- API testing fundamentals
- Selenium automation workflow

10. Recommendations

- Improve UI consistency across browsers
- Fix form validation issues
- Optimize web table performance
- Improve popup handling functionality
- Enhance responsiveness of UI elements

11. Conclusion

This project provided hands-on experience in both manual and automation testing. It helped in understanding real QA processes including test planning, execution, bug reporting, and API

validation. The combination of Selenium automation and Postman API testing enhanced practical knowledge of software quality assurance.

12. Attachments

- Automation Scripts (Selenium)
- Screenshots (Execution + Results)
- Postman API Screenshots
- Test Cases Document
- Bug Report Sheet